Algorithms for Data Analysis and Data Mining

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A. Data Analysis and Regression

1. Logistic Regression

(a)

The analysis of the full dataset was peformed using a logistic regression model:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \cdot hours$$

The threshold for when the function reaches 0.5 and thus the binary output returns 1 was calculated using the following formula.

$$0.5 = \frac{1}{1+e^{-(\beta_0+\beta_1 x)}}$$

$$1 + e^{-(\beta_0+\beta_1 x)} = 2$$

$$e^{-(\beta_0+\beta_1 x)} = 1$$

$$-(\beta_0 + \beta_1 x) = 0$$

$$x = -\beta_0/\beta_1$$

The results show a clear relationship between study hours and the probability of succes. The model provides a good fit to the data and achives classification performance using the chosen decision threshold.

Final model function that resulted:

$$\log\left(\frac{p}{1-p}\right) = -10.83988 + 0.2091565 \cdot hours$$

Final threshold for when the function reaches 0.5 probability:

51.82663

(b)

The analysis of the dataset with last point removed was peformed using the same model and metricts.

$$\log\left(\frac{p}{1-p}\right) = -10.83733 + 0.2091066 \cdot hours$$

Final threshold for when the function reaches 0.5 probability:

51.82679

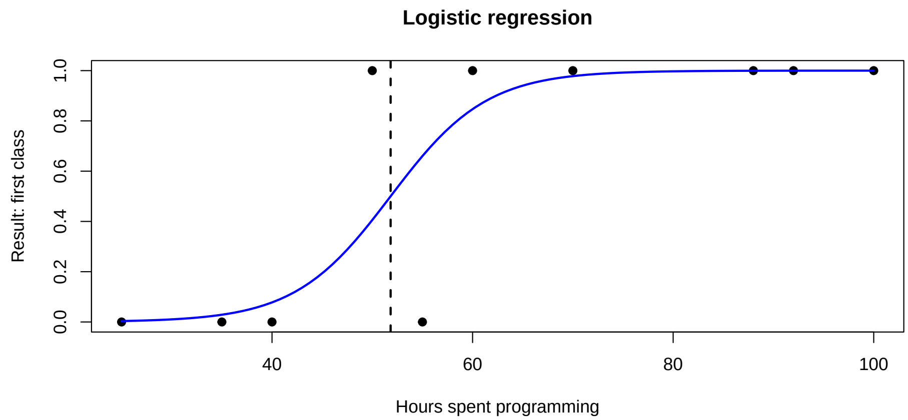

Figure 1: Logistic regression

Removing the point $(100, 1)$ does not significantly change the result. The fitted curve and the threshold remain almost the same. This is because the point $(100, 1)$ is consistent with the general pattern in the data: students with high numbers of hours usually obtained result 1.

The model already has several high-hour observations with result 1, for example 70, 88 and 92 hours. Therefore, the point $(100, 1)$ does not add much new information to the model.

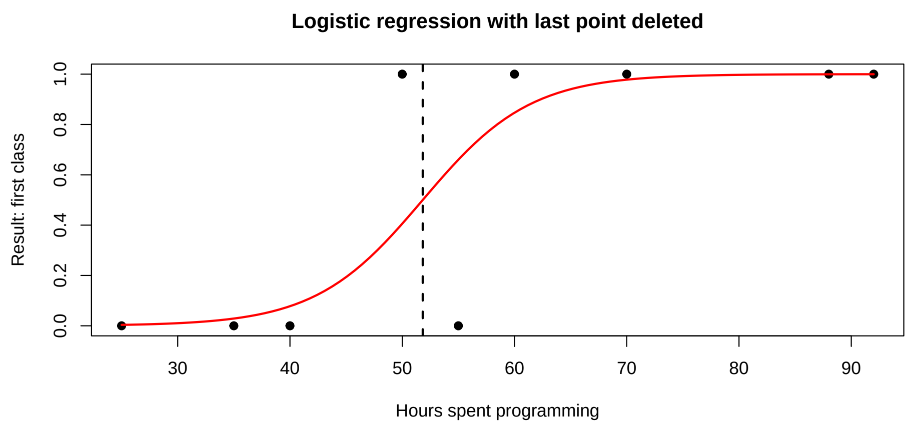

Figure 2: Logistic regression with last point deleted

(c)

The analysis of the dataset with added data point was peformed using the same model and metrics.

$$\log\left(\frac{p}{1-p}\right) = -10.84002 + 0.2091594 \cdot hours$$

Final threshold for when the function reaches 0.5 probability

$$51.82662$$

Adding the point $(115, 1)$ very slightly affects the estimated model. The coefficient associated with the predictor becomes greater, indicating a steeper increase in the probability of success.

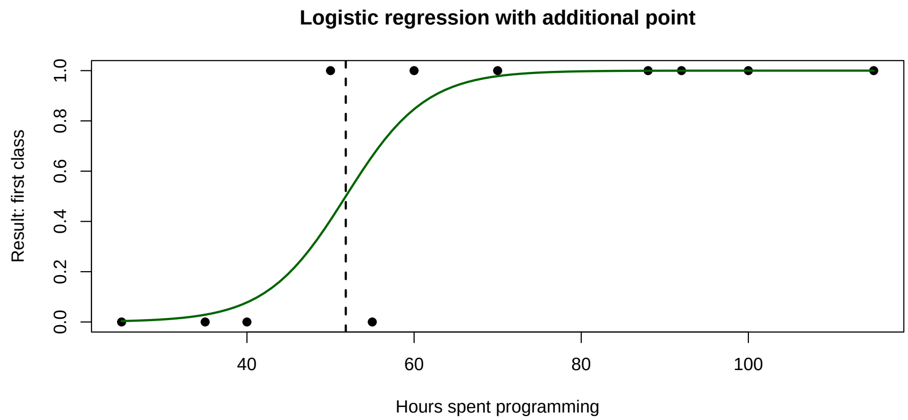

Figure 3: Logistic regression with additional point

Table 1: Comparison of logistic regression models

model	threshold	beta0	beta1
Original	51.8266	-10.8399	0.2092
Without (100,1)	51.8268	-10.8373	0.2091
With (115,1)	51.8266	-10.8400	0.2092

(d)

Logistic regression is a simple classification method for binary outcomes. One of its main advantages is that the estimated coefficients can be interpreted in terms of log-odds and odds ratios, which makes the relationship between predictors and the response variable easier to understand. Another important advantage is that the model produces predicted probabilities in the range:

$$0 \leq p \leq 1$$

making it a valiable method for probabilistic interpretation and decision-making problems. It is also computationally efficient and performs well on relatively small datasets. Lastly, it does not require the predictor variables to follow a normal distribution.

A major limitation of logistic regression is the assumption of a linear relationship between the predictors and the log-odds of the outcome:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x$$

If the true relationship is non-linear, the model may provide poor predictive performance. This method can also be sensitive to outliers and influential observations, especially in small datasets although the addition of the point $(115, 1)$ caused only a small change in the estimated coefficients and decision boundary. Finally, logistic regression may perform worse than more advanced machine learning methods when dealing with complex decision boundaries or large numbers of interacting variables.

2. Nonlinear Least-Squares

(a)

(b)

(c)

(d)

B. Given Dataset

3. Data analysis

We have decided to work on the dataset `sggw_c.csv` and to perform SVM analysis.

(a) Data summary and visualization

Descriptive statistics of the data include mean, median, variance, as well as minimum and maximum. Not all features are necessarly numerical categories, so descriptive statistics should be interpreted with caution, the same applies to the plots below.

variable	min	max	mean	median	variance
Account.Status..with.the.bank.	1	4	2.49	2.00	1.51
Credit.Months	6	60	20.07	18.00	135.10
Credit.Amount..K.1000.	3	159	31.38	21.00	746.60
Saving..account..bonds.	1	5	2.14	1.00	2.44
Years.of.Employment	1	5	3.56	3.00	1.39
Residence.Years	1	4	2.81	3.00	1.19
Age	20	74	36.43	34.00	150.76
House..1.own..2.rent.	1	4	1.44	1.00	0.36
Existing.credit.installment	1	2	1.15	1.00	0.13
Job..1.Good.	1	2	1.48	1.00	0.25
Dependent..e.g...children.	1	2	1.05	1.00	0.04
Credit..1.Good.	1	2	1.28	1.00	0.20

Table 2: Descriptive statistics of the dataset.

Data visualization was performed both in histogram and in boxplot.

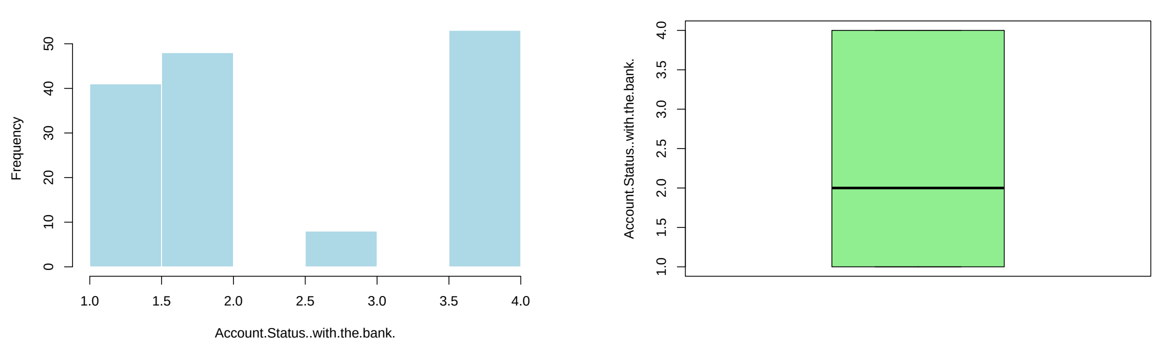

Figure 4: Distribution of Account Status with the bank

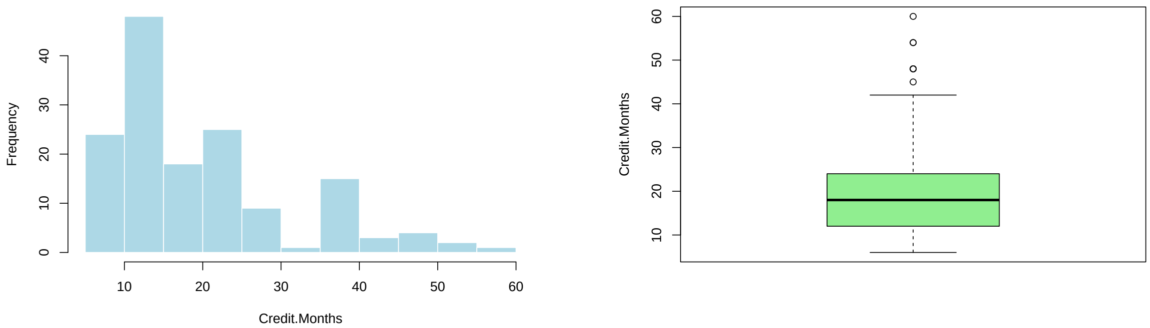

Figure 5: Distribution of Credit Months

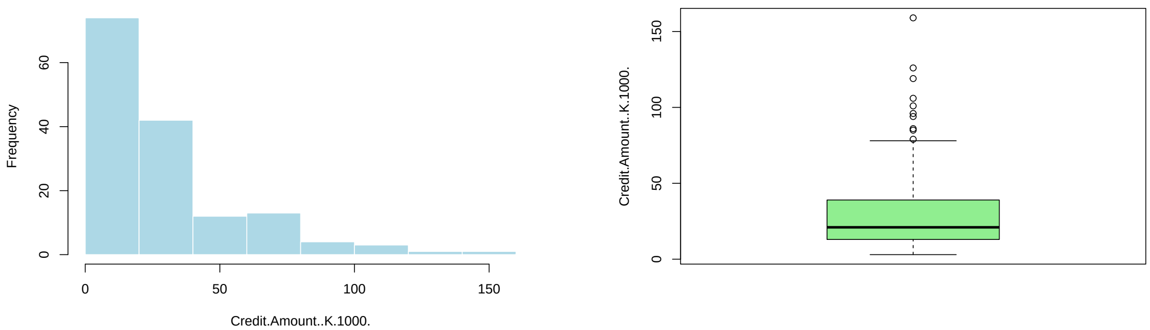

Figure 6: Distribution of Credit Amount

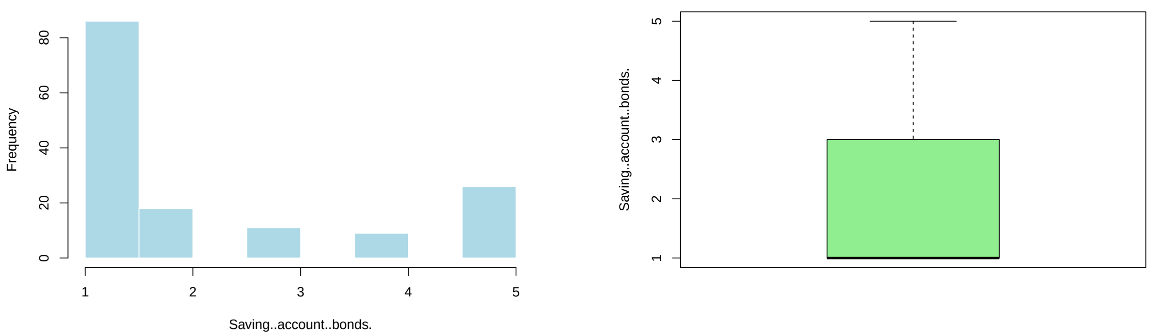

Figure 7: Distribution of Saving account bonds

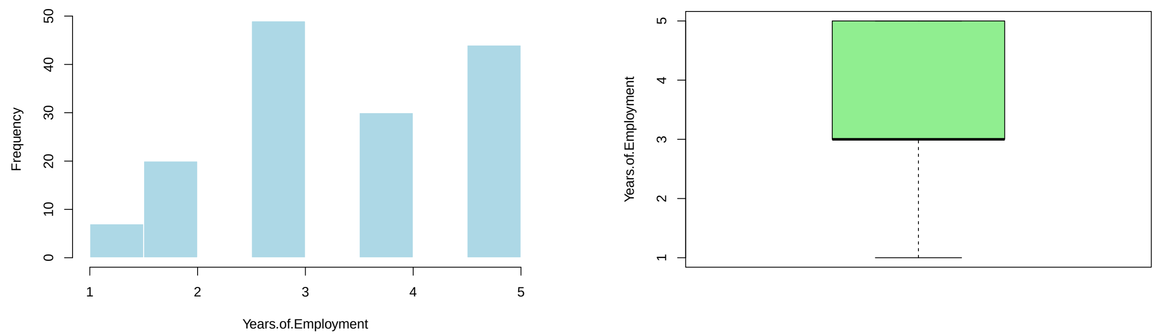

Figure 8: Distribution of Years of Employment

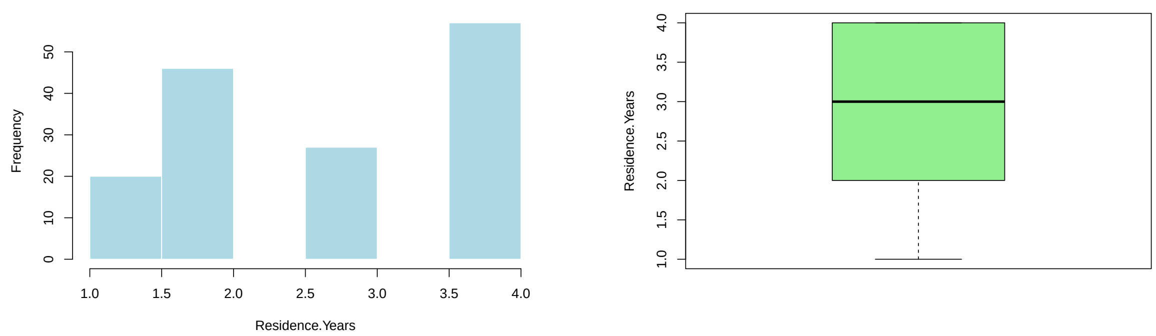

Figure 9: Distribution of Residence Years

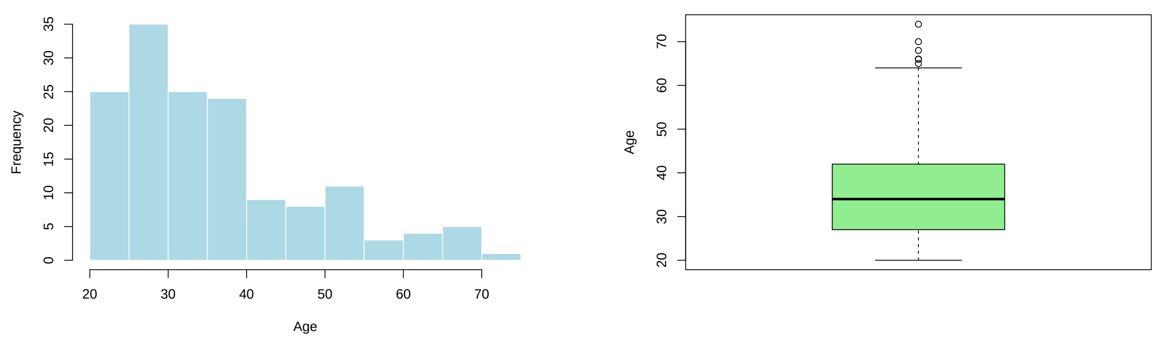

Figure 10: Distribution of Age

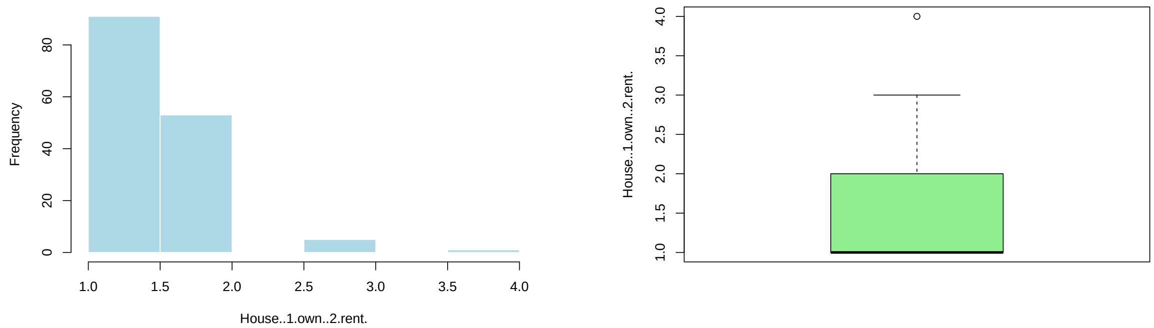

Figure 11: Distribution of House Ownership or Rental

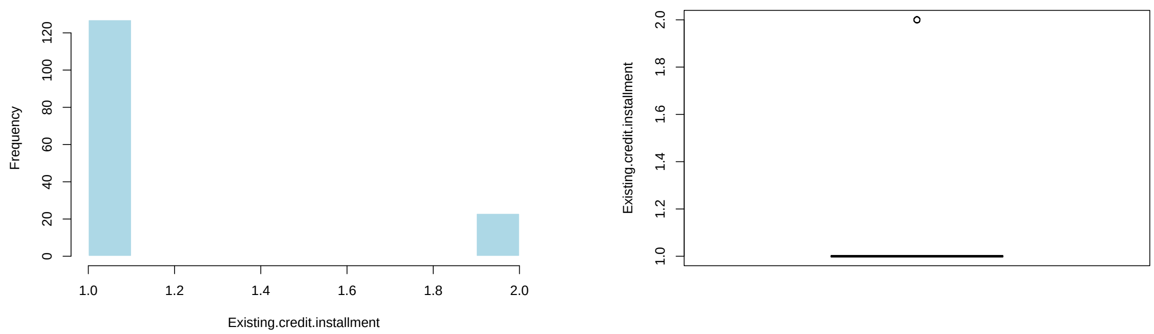

Figure 12: Distribution of Existing Credit Installments

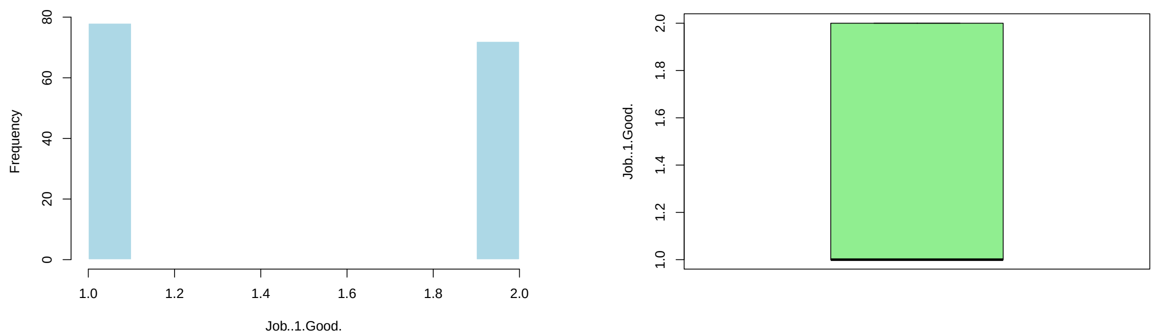

Figure 13: Distribution of Job

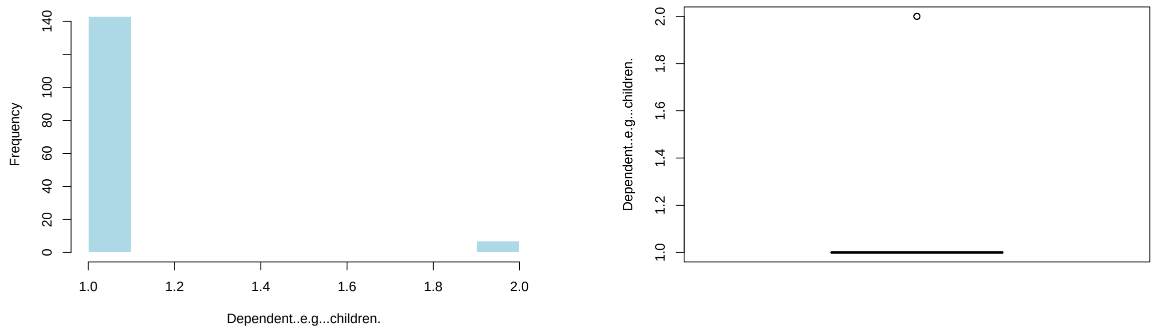

Figure 14: Distribution of Dependent

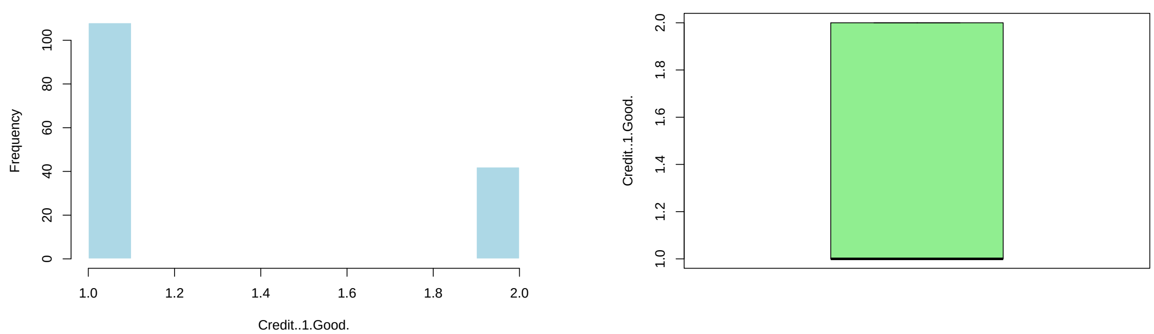

Figure 15: Distribution of Credit

(b) SVM analysis

We have chosen to perform our analysis using Support Vector Machine.

Code is heavily commented but there are a few points worth discussing further.

Sampling the train and test was made using 80 : 20 proportions. Since the amount of input data was limited - 150 entries with 108 `Good` labels and 42 `Bad` labels, we decided to check the proportions of `Good` and `Bad` labels in the sampling split. With the random seed set to 42 the split was as below:

Dataset	class	count	proportion
Full dataset	Good	108	0.72
Full dataset	Bad	42	0.28
Training set	Good	87	0.72
Training set	Bad	33	0.28
Test set	Good	21	0.70
Test set	Bad	9	0.30

Table 3: Class distribution in the full dataset, training set and test set.

As it can be seen, the proportions of the original dataset are similar to the proportions in train and in test datasets, thus the split can be described as representative of the original dataset.

(c) Results

Model SVM trained as below

```
model <- svm(
  x = data_train_x,
  y = data_train_y,
  kernel = "radial",
  scale = TRUE)
```

Resulted in predictions:

Predicted	Good	Bad
Good	21	8
Bad	0	1

Table 4: Confusion matrix for the SVM model.

As we can see, the model works but it performs barely better than a naive model predicting only `Good` label for all cases. It happens because the data is unbalanced, having most of the data labeled as `Good`. Since the problem is credit risk, we would sensibly assume that the more optimal would be a model that makes less mistakes in labeling `Bad` cases as `Good` even if it has an overall lower accuracy.

We can implement it by using class weights which penalize more harshly dalse `Good` predictions.

```
class_weights <- c(
  Good = 1,
  Bad = sum(data_train_y == "Good") / sum(data_train_y == "Bad")
)

model_weighted <- svm(
  x = data_train_x,
  y = data_train_y,
  kernel = "radial",
  scale = TRUE,
  class.weights = class_weights
)
```

Above model on the same train/test split performer as follows.

Predicted	Good	Bad
Good	16	6
Bad	5	3

Table 5: Confusion matrix for the SVM model with weighted classes.

The resulted acuracy is clearly worst, but we have successfully limited false `Good` labels. Depending on the prefered types of errors the weights can be further adjusted. Below are the final metrics of the predictions of both SVM models.

	Accuracy	Precision_Bad	Recall_Bad	Specificity	F1_Bad
Standard_SVM	0.73	1.00	0.11	1.00	0.20
Weighted_SVM	0.63	0.38	0.33	0.76	0.35

Table 6: Comparison of evaluation metrics for standard and weighted SVM models.

C. Mini-Project